

CHILLICOTHE 22 ENE, MO, USA

WMO: 725550

Lat: 39.8668N

Lon: 93.1470W

Elev: 833

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 13301

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.6	5.0	-8.5	3.6	0.0	-4.1	4.5	5.6	18.9	28.6	17.6	25.9	4.7	N/A	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	92.9	76.0	90.0	76.0	87.4	74.5	79.4	88.7	77.8	86.0	76.2	84.5	7.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	144.3	86.1	75.2	136.5	83.9	73.7	129.3	81.6	43.6	89.0	41.8	85.8	40.2	84.6	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
16.3	14.4	12.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-8.3	96.0	5.4	3.5	-12.2	98.6	-15.4	100.6	-18.4	102.6	-22.3	105.1
			-10.0	81.6	5.7	1.9	-14.1	83.0	-17.4	84.2	-20.6	85.2	-24.7	86.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	27.0	29.3	43.3	53.3	63.3	72.9	75.9	74.3	67.5	54.6	42.4	31.1
	DBStd	19.73	12.31	12.40	11.62	9.64	8.33	5.53	5.49	5.31	7.65	9.88	10.36	10.87
	HDD50	2573	714	587	273	74	7	0	0	0	63	267	588	
	HDD65	5496	1177	999	676	361	134	7	2	2	62	348	678	1050
	CDD50	3684	2	8	64	174	420	686	803	753	525	206	40	3
	CDD65	1131	0	0	2	11	82	242	341	290	136	26	1	0
	CDH74	10152	0	0	13	108	629	2170	3346	2552	1126	206	2	0
	CDH80	3573	0	0	0	13	134	731	1322	964	366	43	0	0
Wind	WSAvg	5.7	6.7	6.9	7.4	7.1	5.4	4.9	4.3	4.0	4.4	5.3	5.9	6.0
Precipitation	PrecAvg	39.3	1.3	1.7	2.6	4.0	5.0	5.5	4.7	4.0	4.0	2.9	2.2	1.6
	PrecMax	57.3	2.9	4.1	5.2	6.8	11.8	11.3	15.4	10.6	10.9	7.8	6.9	4.3
	PrecMin	25.3	0.2	0.1	0.5	1.2	1.0	0.8	0.8	0.0	0.7	0.4	0.0	0.0
	PrecStd	8.2	0.7	1.0	1.3	1.5	2.9	2.4	3.3	2.4	2.6	2.0	1.7	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.4	68.5	75.9	81.9	87.8	93.3	97.9	95.9	91.7	84.7	72.1	62.6
		MCWB	50.5	53.0	63.4	66.7	72.3	76.9	73.9	73.1	74.2	70.2	59.3	56.6
	2%	DB	54.0	60.9	71.5	77.3	83.7	89.6	93.2	92.4	87.7	79.7	67.6	55.7
		MCWB	46.4	51.6	59.2	62.8	70.8	75.5	76.8	74.8	73.7	65.9	56.8	49.7
	5%	DB	49.7	53.6	67.0	73.3	80.4	87.2	90.2	88.9	84.8	75.2	62.9	51.0
		MCWB	43.1	47.2	57.0	60.4	68.6	74.2	77.3	74.9	72.1	63.5	54.3	45.9
	10%	DB	44.1	47.3	61.7	69.7	77.5	84.3	87.0	85.4	80.5	70.8	58.5	46.6
		MCWB	39.5	41.3	52.5	58.6	66.4	73.0	75.4	73.7	69.7	61.4	50.6	42.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.4	57.9	65.8	68.5	75.2	79.7	81.4	81.3	76.8	72.3	62.2	59.5
		MCDB	57.8	65.5	74.5	79.3	85.3	89.1	90.6	89.7	88.0	80.4	68.3	63.4
	2%	WB	47.0	51.9	61.4	65.0	72.5	77.9	79.9	78.8	75.1	69.0	58.4	51.8
		MCDB	51.7	58.8	68.8	74.1	81.4	86.7	89.5	86.8	85.6	75.4	64.5	54.8
	5%	WB	42.9	47.3	58.0	62.6	70.5	76.2	78.2	76.6	73.5	66.1	54.7	46.9
		MCDB	48.1	54.5	65.0	71.1	78.7	84.9	86.8	83.9	82.4	71.5	61.6	50.3
	10%	WB	38.5	41.6	53.1	60.1	68.7	74.7	76.7	74.8	71.7	62.5	51.0	42.5
		MCDB	43.2	47.1	59.5	68.4	76.3	83.4	84.6	82.4	78.6	69.3	56.8	46.7
Mean Daily Temperature Range	5% DB	MDBR	17.2	18.1	19.0	20.0	18.6	18.4	18.7	18.9	20.5	20.0	19.0	15.7
		MCDBR	24.1	27.3	24.8	24.9	21.7	21.0	21.1	22.8	22.1	23.8	24.5	21.1
		MCWBR	17.5	18.3	14.8	14.7	11.4	10.2	9.2	9.5	9.4	12.3	16.6	15.6
	5% WB	MCDBR	22.5	25.7	21.0	21.2	19.4	19.0	17.8	18.2	19.2	18.8	21.9	19.1
		MCWBR	16.9	18.7	13.1	14.0	10.8	10.5	9.6	10.0	9.3	12.4	16.4	16.4
Clear-Sky Solar Irradiance	taub		0.295	0.315	0.335	0.392	0.426	0.428	0.441	0.440	0.388	0.345	0.302	0.296
	taud		2.443	2.379	2.404	2.258	2.232	2.259	2.223	2.245	2.342	2.456	2.517	2.464
	Ebn at Noon		278	286	292	280	272	270	265	262	270	272	274	268
	Edh at Noon		25	31	34	42	44	43	44	42	36	29	23	23
All-Sky Solar Radiation	RadAvg		636	911	1192	1540	1794	2004	2007	1755	1510	1036	711	554
	RadStd		69	82	100	108	148	176	131	90	93	115	67	59

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page